

High Temperature Uniaxial Compression and Stress–Relaxation Behavior of India-Specific RAFM Steel



NAIMISH S. SHAH, SAURAV SUNIL, and APU SARKAR

India-specific reduced activity ferritic martensitic steel (INRAFM), a modified 9Cr-1Mo grade, has been developed by India as its own structural material for fabrication of the Indian Test Blanket Module (TBM) to be installed in the International Thermonuclear Energy Reactor (ITER). The extensive study on mechanical and physical properties of this material has been currently going on for appraisal of this material before being put to use in the ITER. High temperature compression, stress–relaxation, and strain-rate change behavior of the INRAFM steel have been investigated. The optical microscopic and scanning electron microscopic characterizations were carried out to observe the microstructural changes that occur during uniaxial compressive deformation test. Comparable true plastic stress values at 300 °C and 500 °C and a high drop in true plastic stress at 600 °C were observed during the compression test. Stress–relaxation behaviors were investigated at 500 °C, 550 °C, and 600 °C at a strain rate of 10^{-3} s^{-1} . The creep properties of the steel at different temperatures were predicted from the stress–relaxation test. The Norton's stress exponent (n) was found to decrease with the increasing temperature. Using Bird–Mukherjee–Dorn relationship, the temperature-compensated normalized strain rate vs stress was plotted. The stress exponent (n) value of 10.05 was obtained from the normalized plot. The increasing nature of the strain rate sensitivity (m) with the test temperature was found from strain-rate change test. The low plastic stability with $m \sim 0.06$ was observed at 600 °C. The activation volume (V^*) values were obtained in the range of 100 to 300 b^3 . By comparing the experimental values with the literature, the rate-controlling mechanisms at the thermally activated region of high temperature were found to be the nonconservative movement of jogged screw dislocations and thermal breaking of attractive junctions.

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I. INTRODUCTION

REDUCED activation ferritic martensitic (RAFM) steel is the most realistic and primary candidate as a breeding blanket structural material for the fusion reactor because of its resistance to radiation–induced swelling and He embrittlement coupled with low thermal expansion coefficient and high thermal conductivity compared with austenitic stainless steel.^[1,2] The RAFM steel is a modified grade of High Chromium (9Cr-1Mo) steel. 9Cr-1Mo steels, finding applications in steam generators, petrochemical industries, and fossil fuel

plants, contain high transmutation nuclei of Mo, Nb, Ni, Ti, *etc.*, which have high residual radioactivity.^[3,4] For easy handling of the disposed blanket, decommissioning of fusion plant, and to avoid harmful effects on environment, it is necessary to develop steels with reduced residual characteristics. RAFM steels are produced by replacing the highly residual radioactive elements such as molybdenum and niobium with low-residual radioactive elements like tungsten and tantalum in the ranges of 1 to 2 wt pct and 0.02 to 0.18 wt pct, respectively.^[5–7]

Development of India-specific RAFM (INRAFM) steel for the Indian Test Blanket Module (TBM), having composition similar to EUROFER 97, was carried out initially to manifest the capability of Indian steel industry in producing the clean steel with stringent control on composition and mechanical properties. The characterizations of physical and mechanical properties were carried out, and it was concluded that the developed steel possessed tensile, creep, impact, low cycle fatigue properties nearly similar to that of EUROFER 97. In the second phase, the content of

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Table I. Chemical Composition (Wt Pct) of the India-Specific RAFM Steel

	Element									
	Cr	C	Mn	V	W	Ta	N	O	P	S
Wt pct	9.03	0.126	0.56	0.24	1.39	0.06	0.03	0.002	0.002	0.001
	Element									
	B	Ti	Nb	Mo	Ni	Cu	Al	Si	Co	As + Sn + Sb
Wt pct	0.0005	0.005	0.001	0.002	0.005	0.002	0.0035	0.06	0.005	0.004

tungsten was varied from 1 to 2 wt pct and tantalum content from 0.06 to 0.14 wt pct with the aim to develop INRAFM steels. The steels were normalized and tempered. The physical and mechanical properties were studied.^[7–9] Steel with 1.4 wt pct W-0.06 wt pct Ta composition was found to possess the optimal combination of properties and was specified as conforming to INRAFM steel.^[7] In the third stage, detailed characterizations of physical and mechanical properties of INRAFM steel are being carried out.

The RAFM steel derives its strength from the tempered martensite microstructure comprising high dislocation density along with $M_{23}C_6$ types of carbides and fine (Ta, V)-carbides.^[7] The tensile strength of RAFM steel decreases with the increasing temperature and increases with the increasing strain rates.^[10–12] The decrease in the tensile strength with the temperature is due to coarsening of the carbides and carbonitrides at higher temperature, which reduces the effective blocking of dislocations and thus decreases the strength. However, the rate of decrease in the strength was minimum in the intermediate range of 450 to 650 K temperature regime. This lower rate of drop in strength has been attributed to the occurrence of the dynamic strain aging.^[7,9,12] The tensile strength increased with the increasing tungsten concentration. This is due to the solid-solution strengthening by tungsten.^[7] Ductility decreased up to 650 K and then increased rapidly beyond 650 K. INRAFM steel possesses tensile strength and ductility comparable to those internationally developed EUROFER 97, F82H, and CLAM RAFM steels.^[7,13,14]

The creep behavior of the steel showed a small instantaneous strain on loading, a relatively brief primary stage, and an apparent secondary stage followed by a prolonged tertiary stage.^[7,9,15] The minimum creep rate increased with the increasing temperature and stress and obeyed the Norton's power law with high stress exponent (n) values. The stress exponent was found to decrease with the increasing temperature.^[9,16,17] The creep rupture life of INRAFM steel has been better than the internationally developed RAFM steels.^[7]

It is very important to understand the plastic flow behavior of structural material to predict the performance of the nuclear reactor. Flow stress is the important factor to evaluate the plastic flow properties.^[18] Also, the direct measurement of creep properties

can be time consuming, and the number of experiments required to generate the creep data are also more. In comparison, the stress-relaxation test can be effectively utilized to determine the creep properties of material.^[19] Krishna *et al.*^[20] observed a certain difference in the creep properties estimated from stress-relaxation test to that obtained from long time creep test. It was concluded that the stress-relaxation test for alloys with prominent primary creep regime cannot be useful for estimating and may overestimate the creep rate. However, stress-relaxation test can be used for low alloy ferritic steels as a brief primary stage has been observed for these alloys.^[9]

In this study, the high temperature uniaxial compression and stress-relaxation tests on the INRAFM steel were conducted. The temperatures were chosen to simulate the working conditions in the reactor. The stress-relaxation behavior of the steel was analyzed, the creep properties were estimated, and the rate-controlling mechanism was determined from activation volume (V^*).

II. EXPERIMENTAL

A. Material

The RAFM steel was produced by M/s. Mishra Dhatu Nigam Limited (MIDHANI), Hyderabad, India. The chemical composition of the steel is shown in Table I. To obtain the steel of precise composition and homogeneity in structure, vacuum induction melting route followed by vacuum arc refining route was followed. Hot forging and hot rolling were carried out in the temperature range from 1400 to 1500 K for breaking the cast structure. The material was subjected to heat treatment at 1253 K for 30 minutes followed by water quenching to achieve martensitic structure. Finally the steel was given tempering treatment at 1033 K for 90 minutes followed by air cooling.

B. Testing Procedure

Cylindrical test specimens with 5 mm diameter and 8 mm length were machined and used for both hot compression and stress-relaxation tests. The uniaxial isothermal hot compression tests were conducted on the Zwick Roell 100 kN screw-driven universal testing machine. A resistance heating split furnace with kanthal

heating element was used to heat the specimen. Thermocouple of K-type chromel alumel was used to measure the specimen temperature. The specimens were coated with high temperature lubricant moly-disulphide to minimize friction between specimens and die during the experiment. The specimens were heated to test temperature at a heating rate of 10 °C/min and then soaked at that temperature for 30 minutes. The test temperature was maintained within ± 5 °C. The compression tests were conducted at 25 °C, 300 °C, 500 °C, and 600 °C. All the tests were carried at a constant strain rate of $1 \times 10^{-3} \text{ s}^{-1}$. Standard equations were used to convert the load–displacement data in to true stress–true strain data. The elastic region was subtracted from true stress–true strain curve to obtain true plastic strain for each temperature.

The stress–relaxation tests were conducted at 500 °C, 550 °C, and 600 °C temperatures. The specimens were deformed to 10 pct strain and held for 1 hour. All the tests were carried at a constant strain rate of $1 \times 10^{-3} \text{ s}^{-1}$. The decrease in the compressive stress was measured during the stress–relaxation. The specimens were again reloaded for 4 pct strain and unloaded to obtain the unloading compliance (K). The specimens were again loaded to perform strain-rate change. To analyze the stress–relaxation data for estimating creep properties, true stress *vs* relaxation time was plotted to find the stress rate ($\dot{\sigma}$). Machine compliance was found by taking slope of unloading part. The stress-rate data were obtained from stress–relaxation test and were converted to strain rate using the relation $\dot{\epsilon}_{\min} = \frac{\dot{\sigma}}{K}$. Strain rates thus obtained from stress–relaxation data correspond to the minimum creep rates obtained from the creep tests.^[19,21] The minimum creep rate ($\dot{\epsilon}_{\min}$) *vs* stress (σ) was plotted to find the Norton's stress exponent value (n).

The two different specimens were annealed at 500 °C and 600 °C to observe the effect of isothermal holding on the microstructure.

C. Microstructure Characterization

The deformed specimens were cut along the longitudinal axis. Metallographic processes were carried out on the as-received specimen and uniaxially compressive deformed along with annealed specimens to study the microstructural features. The etchant used was freshly prepared Vilella's reagent (1 g picric acid, 5 ml HCl, 100 ml ethanol). Optical microscopy and field emission-scanning electron microscopy (FE-SEM), and the electron backscattered diffraction (EBSD) investigations were carried out to study the microstructural details of the specimens. For EBSD scan samples were electropolished using an electrolyte made of 20 pct perchloric acid and 80 pct methanol. Electropolishing was carried out at -40 °C using 20 V for 20 seconds.

III. RESULTS AND DISCUSSION

The optical micrograph of the as-received normalized and tempered RAFM steel has been shown in Figure 1(a). The micrograph shows the tempered

martensite structure consisting of martensite laths arranged in prior austenitic grains. Figure 1(b) shows the EBSD inverse pole figure (IPF) map of the as-received sample. It is evident from the IPF map that the steel has heterogeneous and complex grain structure which is typical of tempered martensitic steel. Grain size distribution obtained from the EBSD data is shown in Figure 1(c). Average grain size was $5.5 \pm 1.6 \mu\text{m}$. Figure 1(d) shows the (001) pole figure of the as-received steel indicating random texture of the as-received material. The average hardness of as-received specimen was $225 \pm 5 \text{ H}_V$.

The true plastic stress–strain curve for RAFM steel under uniaxial compression at different temperatures is shown in Figure 2. The true stress–strain curves are obtained by converting load–displacement data using standard equations and are corrected for removing barrelling effect on the specimen during deformation. Rapid increase in stress at lower strain followed by steady state behavior is seen. The increase in stress with the increasing strain for room temperature deformation can be attributed to work hardening phenomenon occurring in the material.^[18] The decrease in the peak stress level is observed with the increasing temperature which shows the temperature sensitivity of INRAFM steel.^[12] The similar dependence of flow stress on temperature for CLAM steel was reported by Wang *et al.*^[18] The variation of the yield strength with the temperature for INRAFM steel is shown in Figure 3. The increase in the flow stress (Figure 2) and the presence of the plateau region (Figure 3) in the temperature range of 300 to 500 °C might be due to occurrence of the dynamic strain aging (DSA).^[12] Similar behavior was observed for other internationally developed low-activity ferritic martensitic steel.^[18]

With the increasing temperature, the yield strength decreased up to 300 °C due to the thermally activated nature of deformation. In the intermediate temperature range (300 to 500 °C), the yield strength remained almost unaltered due to the hindrance caused by the solute atoms (DSA effect) to dislocation movement.^[12] At high temperatures, above 500 °C (beyond the DSA regime), INRAFM steel exhibited a decrease in the yield strength values with the increasing temperature. The perceptible decrease in the strength can occur due to (i) softening due to coarsening of the precipitates at high temperatures during test and (ii) softening due to occurrence of dynamic recovery.^[12] This decrease in the strength is in good agreement with the behavior of many other ferritic–martensitic steels.^[5,13,14,22]

The stress–relaxation test behavior of the INRAFM along with the strain-rate change test for determining the strain-rate sensitivity (m) at 550 °C is shown in Figure 4. Similar behaviors were observed at 500 and 600 °C. The m values at 500, 550, and 600 °C are 0.018, 0.04, and 0.06, respectively, and a subtle increment was found with the increasing temperature (Figure 5). The lower value of m denotes the flow instability which may result in localized deformation leading to fracture.^[23]

The release of compressive stresses ($\Delta\sigma$) over the time for 1 h after holding the cross-head movement at 10 pct compression strain is shown in Figure 6. The

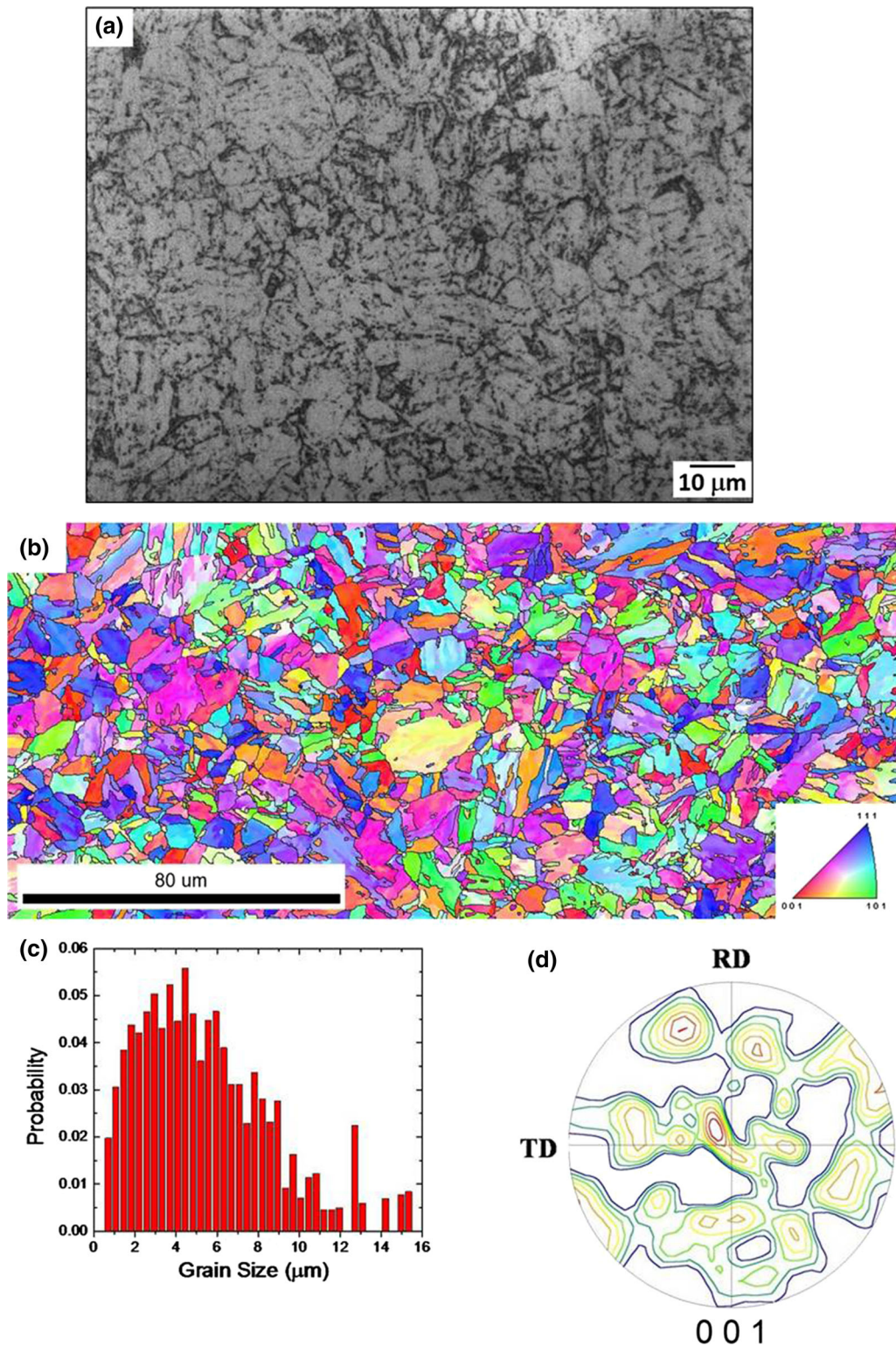


Fig. 1—(a) Optical microstructure of the as-received RAFM steel in normalized and tempered condition. (b) EBSD IPF map of the as-received RAFM sample, (c) Grain size distribution plot. (d) (001) pole figure.

exponential decrease in the stress is observed with time during the test. The stress rate is found to be higher in the initial period of the relaxation. The compression of 10 pct induced the compressive stresses of 452, 383, and 325 MPa at 500 °C, 550 °C, and 600 °C, respectively.

Complete stress-relaxation was observed at 600 °C. The stress dropped to 1 MPa after 30 minutes and remained steady thereafter. The stress value dropped to 241 and 132 MPa, at 500 °C and 550 °C, respectively. Rapid stress-relaxation is indicative of degradation of creep

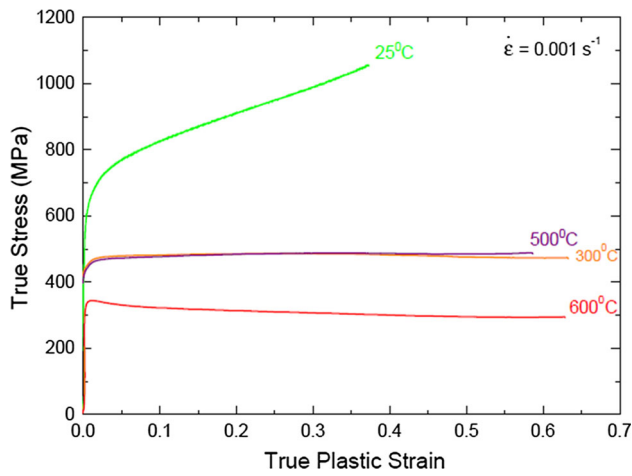


Fig. 2—Uniaxial compression plastic behavior of INRAFM steel.

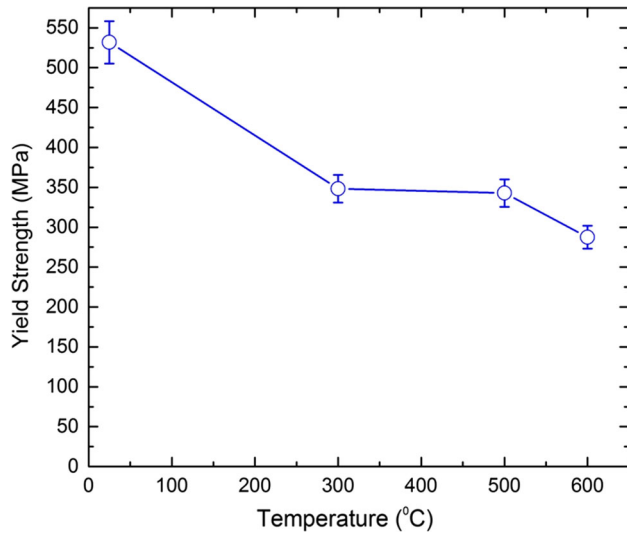


Fig. 3—Variation of yield strength with temperature for INRAFM steel.

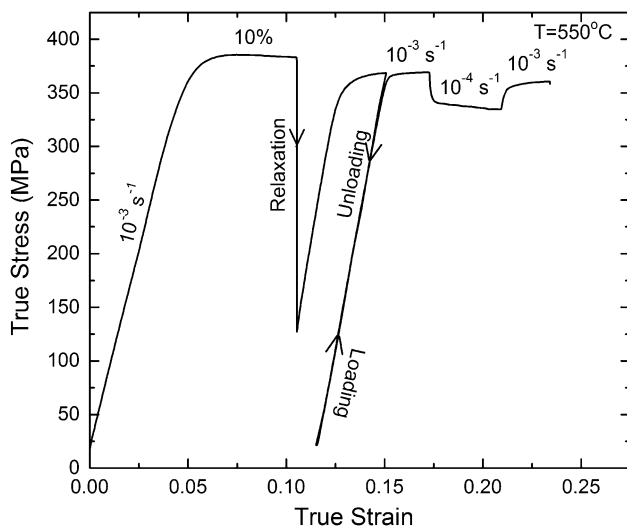


Fig. 4—Typical stress-relaxation and strain-rate change behavior of INRAFM steel at 550 °C.

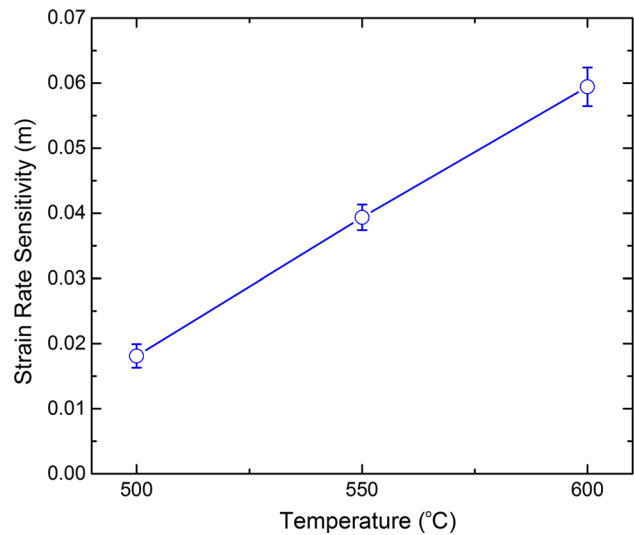


Fig. 5—Variation of strain-rate sensitivity with temperature for INRAFM steel.

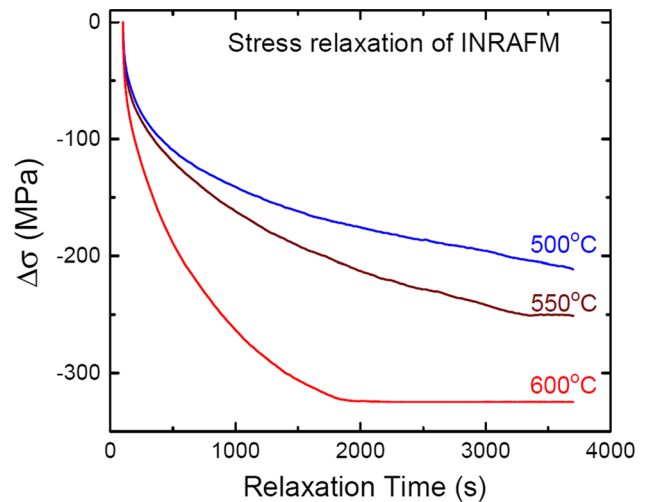


Fig. 6—Variation in $\Delta\sigma$ with relaxation time for INRAFM steel at various temperatures.

properties of the steel at 600 °C. Similar degradation in the creep strength at 550 °C for long testing time was observed for EUROFER'97 steel.^[16,23,24] It was concluded that the precipitation of M_6C -type precipitates rich in tungsten (W) might be the reason for the decrease in the strength. Strength of low-activity ferritic–martensitic steel is contributed by the solid-solution strengthening effect of W.^[7,9,24] The formation of W-rich M_6C precipitates results in the depletion of W in the matrix. The decrease in W content results in loss of solid-solution hardening. Similar precipitation of M_6C precipitates at 550 °C has been observed.^[25]

Creep rates ($\dot{\epsilon}_{min}$) at different true stresses (σ) for a particular temperature are derived from the stress–relaxation data using the formalism described in the Section II–B. The variations of $\dot{\epsilon}_{min}$ with σ for different temperatures are shown in Figure 7. The variation of the $\dot{\epsilon}_{min}$ with σ is described by the Norton's law, power

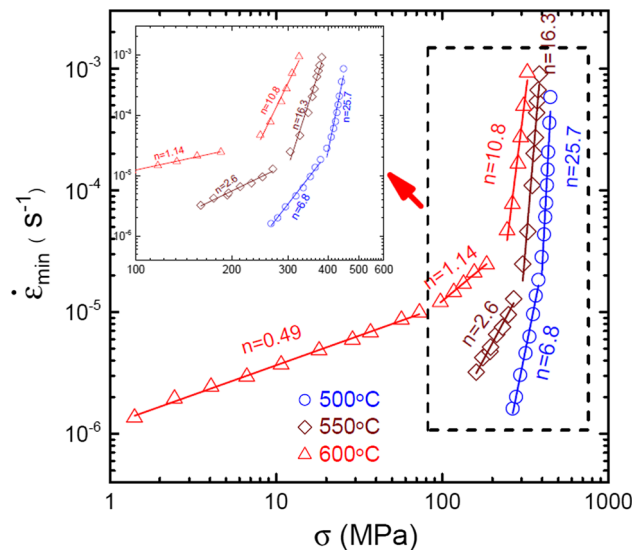


Fig. 7—Variation of minimum creep rate with true stress of the INRAFM steel at different test temperatures.

relation of the form: $\dot{\epsilon}_{\min} = k\sigma^n$, where k is a constant and n is Norton stress exponent. The increase in the minimum creep-rate value with the increasing stress and temperature is observed, as shown in Figure 7. The stress exponent (n) is a function of temperature and stress. It decreases with the increasing temperature and increases with the increasing stress (Figure 7). The decreasing behavior of n with temperature has also been observed for various low-activity ferritic–martensitic steels.^[9,16,17,26] This decreasing behavior of n is attributed to the coarsening of $M_{23}C_6$ - and MX-type precipitates.^[10,27] Fernandez *et al.*^[16] studied the variation of minimum creep rate with stress for the EUROFER steel, and similar stress exponent values were observed in the 500 to 600 °C temperature region. The coarsening results in incoherent precipitates which reduce the coherency strains between precipitates and matrix, thereby reducing the stress exponent. Creep deformation characteristics are governed by Norton's law as the change in the stress exponent changes the creep deformation mechanism. Figure 7 shows change in slopes of the data points at lower stresses for a particular temperature. These types of changes in slope have also been observed in other alloys and are indicative of transition in creep mechanism with stress.^[28] In general, the stress exponent is determined for higher value of stress, and it is extrapolated for lower stress values. This underestimates the creep-rate value and may predict false creep deformation mechanism. This urges in studying the variation of creep rate at lower stress. As seen in Figure 7, the power law is obeyed only in the central region of stress. At lower stress, the linear dependence on the stress is seen and is known as Harper–Dorn or Nabarro–Herring or Coble creep.^[29] Creep mechanisms at lower stresses are in general diffusion based. These mechanisms lead to higher creep rates compared with the dislocation-based mechanism at comparable stresses. Stress–relaxation tests are very

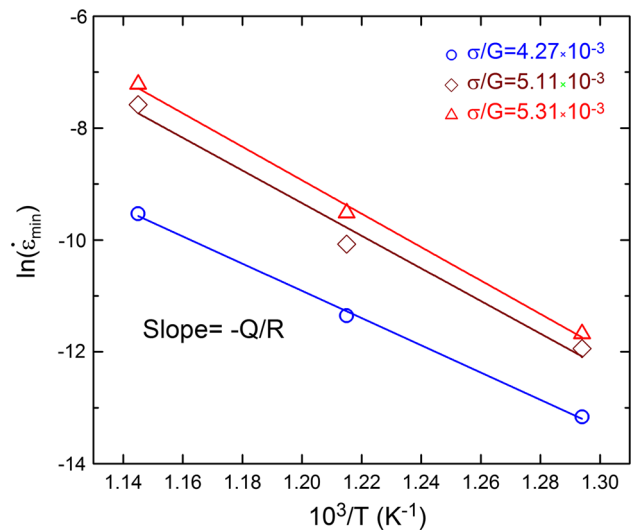


Fig. 8—Arrhenius plots for calculating the activation energies at different modulus-compensated stresses.

useful in generating creep data at lower stresses within reasonable time.

According to the equation $\dot{\epsilon}_{\min} = A_0\sigma^n \exp(-\frac{Q_c}{RT})$, the activation energy (Q_c) can be calculated^[30] from the slope of a semilogarithmic plot of the strain rate and the inverse of temperatures at various modulus-compensated stress rates (Figure 8). The average activation energy was found to be $\sim 241 \pm 25$ KJ/mol. The activation energy for self diffusion (Q_d) for α -iron is 251 KJ/mol.^[31] The similarity of the Q_c value to Q_d indicates diffusion-based mechanism to be the responsible mechanism of creep in the RAFM steel.

Flow stress values obtained for 10 pct strain at different temperatures (500 to 600 °C) were normalized using Bird–Mukherjee–Dorn relationship: $\dot{\epsilon}_{\min} = \frac{ADGb}{KT} (\frac{\sigma}{G})^n \exp(-\frac{Q_c}{RT})$, where A is the constant, K is Boltzmann's constant, T is the test temperature (K), D is the diffusion coefficient, G is the shear modulus, and b is the Burgers vector.^[30] A bilogarithmic plot of normalized creep rate ($\frac{\dot{\epsilon}_{\min}KT}{DGb}$) vs normalised ($\frac{\sigma}{G}$) is shown in Figure 9. The lattice diffusion coefficient, Burgers vector, and shear modulus were taken of α -ferrite, as the INRAFM steel in the testing temperature range (500 to 600 °C) will be in ferritic phase. Lattice diffusivity D_L (m^2/s) for iron in ferrite at a given temperature was obtained from $D_L = 2.8 \times 10^{-4} \exp(-17T_M/T)$, where $T_M = 1803$ K is the melting point for the low-activity ferritic martensitic steel.^[31,32] The shear modulus G at temperature T was calculated from equation: $G = G_0[1 + \{(T - 300)/T_M\} * (T_M/G_0)(D_g/dT)]$ with $G_0 = 8.1 \times 10^4$ MPa, and $(\frac{T_M}{G_0})(\frac{D_g}{dT}) = -0.81$.^[32,33] The magnitude of Burgers vector b for α -ferrite = 2.48 Å. A good temperature-compensated superimposition of minimum creep rates at different temperatures into a single line with slope (stress exponent) equal to 10.05 is obtained (Figure 9). The stress exponent value obtained from the normalized plot from stress–relaxation test for

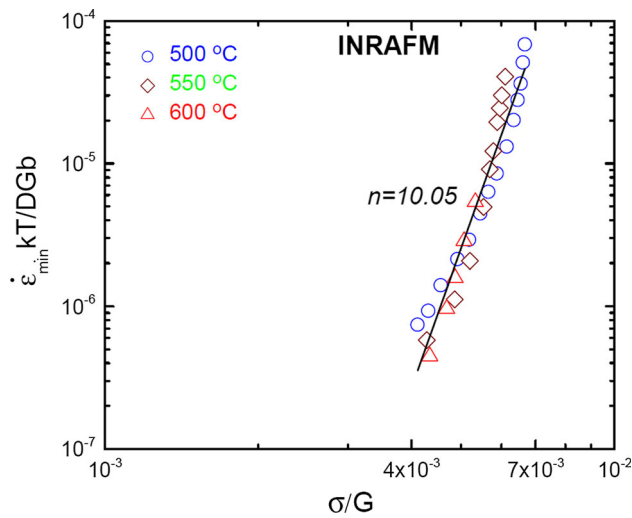


Fig. 9—Variation of normalized strain rate with normalized stress of INRAFM steel.

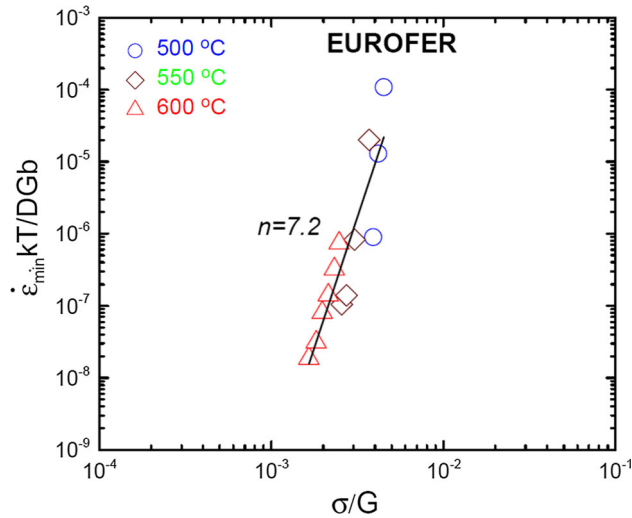


Fig. 10—Variation of normalized strain rate with normalized stress of EUROFER steel (replotted from Ref. [16]).

INRAFM is in good agreement with that obtained from the creep test studied by Mathew *et al.*^[9]

The bilogarithmic plot of the normalized creep rate vs the normalized stress for EUROFER steel is shown in Figure 10. The stress exponent value is higher for INRAFM steel compared with EUROFER steel signifying better creep properties of INRAFM steel. Laha *et al.*^[7] observed the improved creep properties of INRAFM steel in comparison with internationally developed low-activity ferritic–martensitic steels. The improved creep properties of INRAFM steel is inferred to be due to the relative higher content of tungsten. Tungsten retards the recovery of the dislocation and subgrain structure and gets dissolved in $M_{23}C_6$ -type precipitates and reduces the coarsening characteristics, thus reducing the creep rate.^[7]

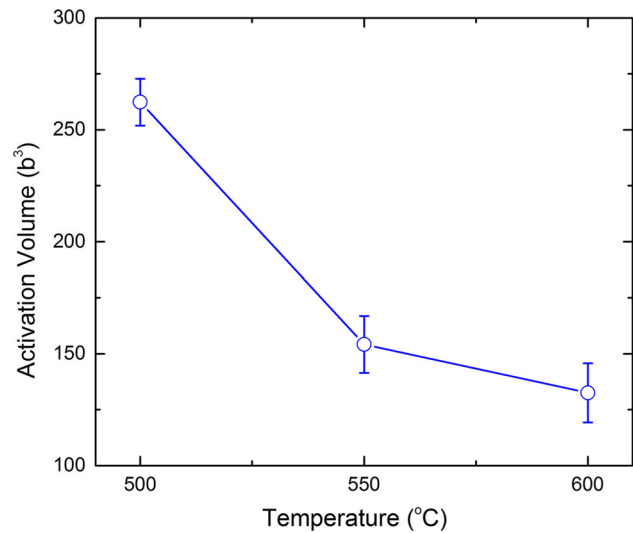


Fig. 11—Variation of activation volume with temperature of INRAFM steel.

The plastic flow in the material at higher temperatures may occur due to diffusion-assisted processes such as cross slip or climb movement of edge dislocations or nonconservative movement of jogged screw dislocations or by thermal breaking of attractive junctions. In principle, the rate-controlling deformation mechanism can be identified by comparing the magnitude of the experimentally determined activation volume (V^*) with the theoretical values predicted by models of thermal-activated flow. The variation of activation volume (V^*) with temperature is shown in Figure 11. The activation volume was estimated from the relationship $V^* = MKT/m\sigma$,^[28] where M is the Taylor factor ($= 3$), K is the Boltzmann's constant, T is the test temperature (K), m is the strain-rate sensitivity, and σ is the stress corresponding to the change in strain rate in strain-rate test to determine m . The activation volume was found to be in the range of 100 to 300 b^3 at all the test temperatures. By comparing this value with literature,^[34] the rate-controlling mechanisms were found to be nonconservative movement of jogged screw dislocations and the thermal breaking of attractive junctions.^[35]

Figure 12 shows the optical micrographs of uniaxial compressive-deformed specimens at different test temperatures. The grains are seen elongated in the deformation direction. Figure 13 shows the scanning electron microscopy images of deformed specimens at 500 °C and 600 °C. The arrows in the black shows the deformation direction. The grains are elongated in the deformation direction. The tempered martensite laths are seen along the ferrite grains. The growth of ferrite grains is observed at 600 °C.

The optical micrographs of the separately annealed specimens at 500 °C and 600 °C with 30-minutes soaking time is shown in Figure 14. The equiaxed grains are seen in the microstructure. Not much difference is seen between the microstructures of the two annealed specimens. Figure 15 shows the scanning electron images of the annealed specimens at 500 °C and 600 °C. The

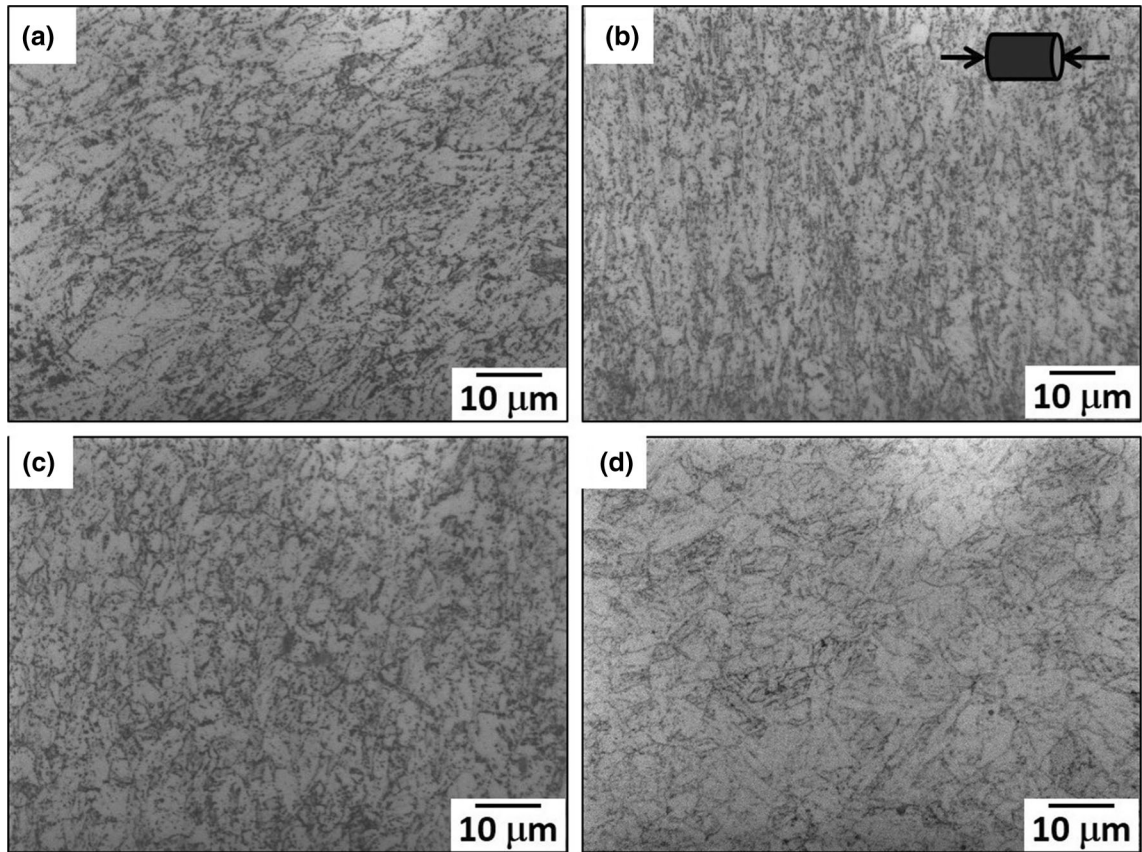


Fig. 12—Optical micrographs of uniaxial compressive-deformed INRAFM steel at (a) 25 °C, (b) 300 °C, (c) 500 °C, (d) 600 °C.

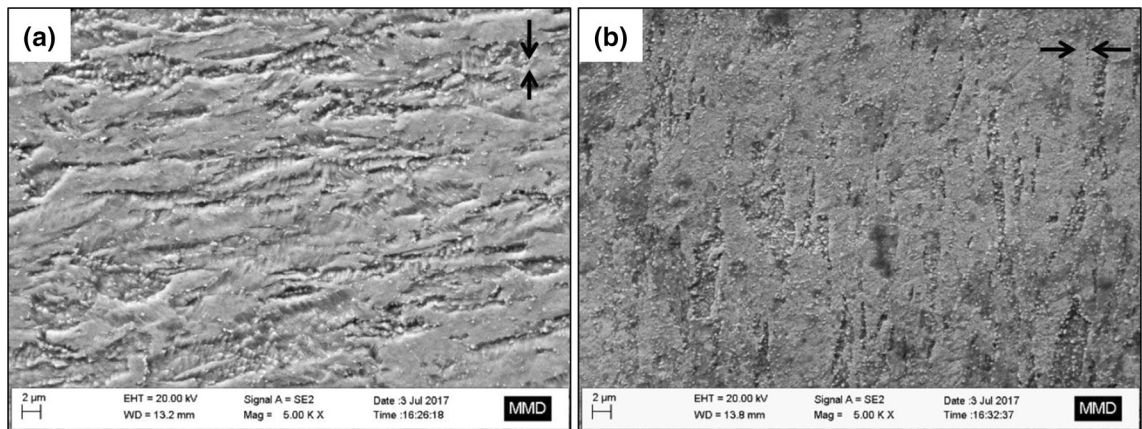


Fig. 13—SEM images of deformed specimens at (a) 500 °C, (b) 600 °C.

tempered martensite laths are seen along the ferrite grains. Fine precipitates are seen in specimens annealed at 500 °C along the ferrite grain boundaries. These precipitates are not visible in the 600 °C specimen. The IPF map and the grain size EBSD distribution obtained from the EBSD data of the sample annealed at 600 °C are shown in Figure 16. The microstructure and grain

size distribution of the annealed sample are similar to the as-received sample (Figure 1(b), (c)). Average grain size of the sample annealed at 600 °C is $\sim 5.9 \pm 1.8 \mu\text{m}$ which indicates that ferrite grain size are stable up to 600 °C. Thus, dissolution of precipitates might be the reason for the drop in mechanical properties of INRAFM steel at 600 °C.

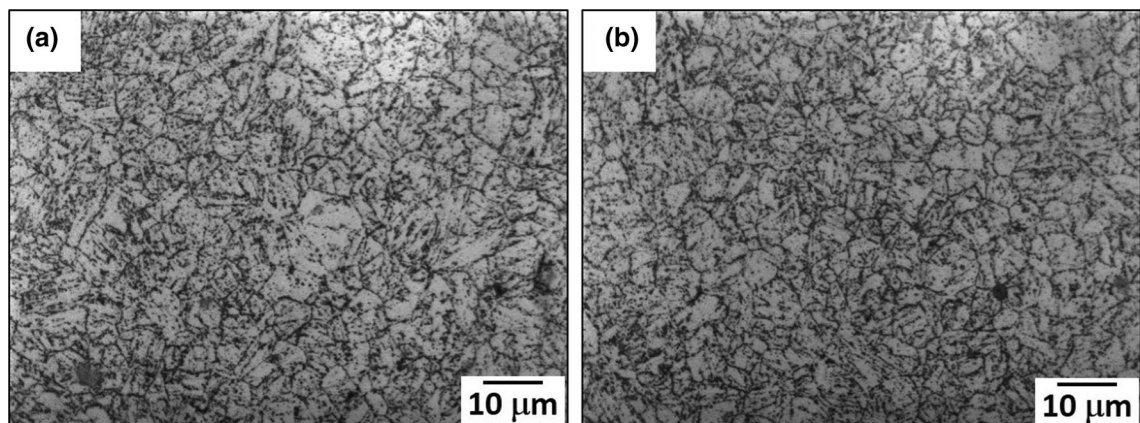


Fig. 14—Optical micrographs of annealed INRAFM steel at (a) 500 °C, (b) 600 °C.

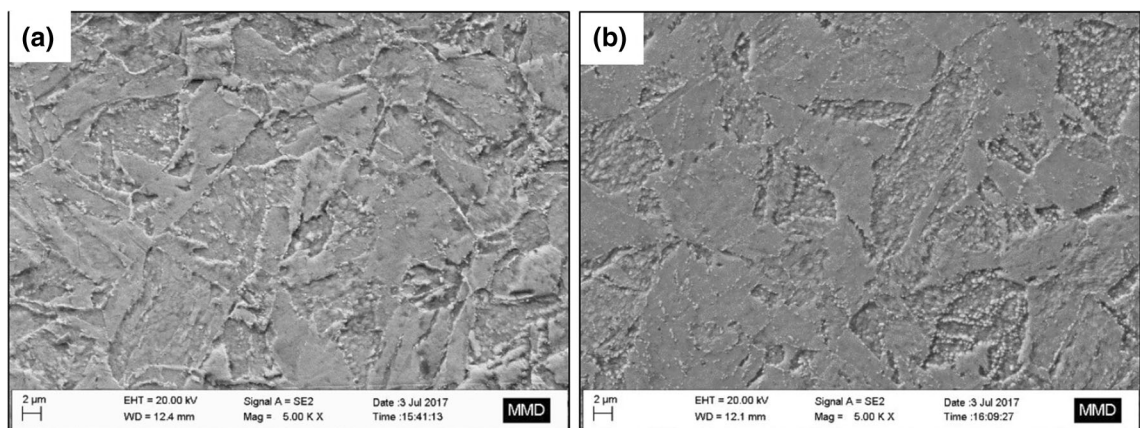


Fig. 15—SEM images of annealed specimens at (a) 500 °C, (b) 600 °C.

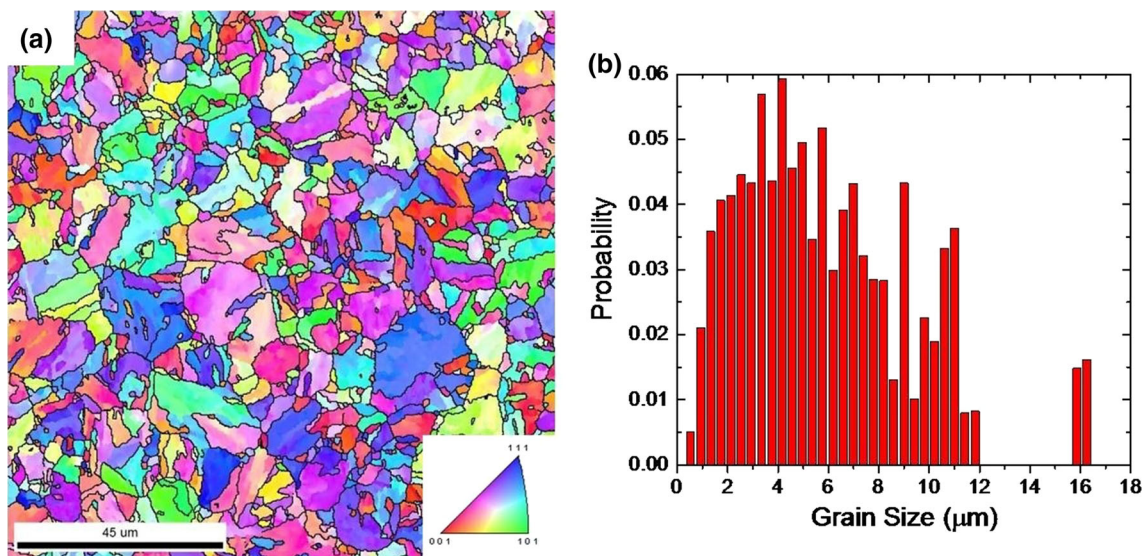


Fig. 16—EBSD (a) IPF map and (b) grain size distribution of the sample annealed at 600 °C.

IV. CONCLUSIONS

- (1) The true stress–true plastic strain curves obtained from the hot uniaxial compression of INRAFM steel exhibited the drastic drop in strength at 600 °C. The strength of INRAFM steel at 500 °C was slightly higher than 300 °C which might be due to dynamic strain aging.
- (2) Strain-rate sensitivity (m) exhibited a subtle increasing behavior with temperature. The value of m at 600 °C was found to be 0.06. The lower value of m up to 600 °C denotes the flow instability of the INRAFM steel. This may result in localized reduction of cross section eventually leading to ductile fracture.
- (3) The creep properties of INRAFM steel obtained from stress–relaxation test at the higher stress regime were found to be comparable with uniaxial creep test properties. At lower stress regime, data obtained from the stress–relaxation tests clearly indicated a change in the creep mechanism in the steel at lower stresses.
- (4) The creep properties of INRAFM steel were found to be better than EUROFER steel. This is inferred to be due to the higher content of W in INRAFM steel which reduces the coarsening characteristics of precipitates thus decreasing the creep rate.
- (5) The temperature-compensated stress exponent (n) value obtained from the normalized plot of strain rate vs stress was 10.05. From the activation volume (V^*), the rate-controlling mechanisms were concluded to be nonconservative movement of jogged screw dislocations and thermal breaking of attractive junctions.

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